TARUN REDDY THANDU

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SUMMARY

AI / Software Engineer (MS in AI, Northeastern, GPA 3.96) who builds production systems end to end: Python, Java, and TypeScript services on PostgreSQL, React and Angular front ends, and the LLM layer on top, including **multi-agent orchestration** (LangGraph, MCP), RAG, guardrail models at **99.7% detection**, and OpenTelemetry observability. Ships on AWS, Azure, and GCP with Docker, Kubernetes, and CI/CD.

TECHNICAL SKILLS

Languages	: Python, Java, TypeScript, JavaScript, SQL, C, Bash
AI / LLM	: LangGraph, MCP, LiteLLM, RAG, multi-agent systems, QLoRA, RLHF, RAGAS, LLM-as-a-Judge
ML / DL	: PyTorch, TensorFlow, Hugging Face, BERT, FAISS, Pinecone, pruning, quantization
Backend & Web	: FastAPI, Spring Boot, Django, React, Angular, REST APIs, Celery, browser extensions
Data	: PostgreSQL, MongoDB, ClickHouse, Redis, Neo4j, Kafka, Pinecone
Cloud & DevOps	: AWS, Azure, GCP, Docker, Kubernetes, Terraform, Jenkins, CI/CD, OpenTelemetry

EXPERIENCE

PointGuard AI | AI / ML Engineer **Aug 2025 - Present**

Python, TypeScript, LangGraph, MCP, FastAPI, PostgreSQL, ClickHouse, Azure, AWS *San Francisco, CA*

- Building **Agent Mission Control**, endpoint security for the agentic era: a scanner that sweeps enterprise Windows fleets to inventory AI agents, tools, skills, and MCP servers, then flags prompt injection, malicious scripts, dangerous access, and data exfiltration.
- Shipped a **cross-browser extension** in TypeScript (WXT, MV3) with per-site adapters that intercept prompts and file uploads on ChatGPT, Claude, and Gemini, rolled out to enterprise pilots through **Microsoft Intune**.
- Engineered **stateful multi-agent orchestration** (LangGraph + MCP servers) for retrieval, policy decisioning, and tool-calling, shipped as a **LiteLLM proxy plugin** with zero-downtime cutover and PII redaction in the agent pipeline.
- Trained and evaluated **guardrail models** to **99.7% detection at a 2% false-positive rate** against **OWASP LLM Top 10** categories.
- Built the **FastAPI + SQLAlchemy (async)** backend over **PostgreSQL** on AWS RDS, with Alembic migrations and a two-role security model enforcing row-level security for multi-tenant isolation.
- Built a **high-throughput LLM observability pipeline** (OpenTelemetry to ClickHouse) capturing prompt traces and agent decisions, with monitoring dashboards and real-time anomaly alerting for production.
- Built a **four-tier classification pipeline** (fleet telemetry, known-list fast path, network interception, LLM classification) that detects shadow AI, publishing a knowledge base of **30,000+ AI sites scored by risk**.

HER Heard | AI / ML Engineer Intern **Sep 2024 - Dec 2024**

LangGraph, React, FastAPI, Redis, Pinecone, Neo4j, GCP *Boston, MA*

- Built an **agentic medical Q&A bot**: LangGraph retrieval, synthesis, and validation agents on React/FastAPI/Redis at **sub-300ms p95**.
- Built **hybrid retrieval** (Pinecone plus a Neo4j knowledge graph), reaching **65% accuracy** on multi-hop queries.
- Fine-tuned **LLMs (Llama) with QLoRA** and built an **RLHF pipeline** on physician-validated preference data, improving safety **42%**.
- Built the **evaluation layer**: RAGAS faithfulness at **78%**, Constitutional AI guardrails, and fact-checking agents that catch hallucinations.

Infosys | Specialist Programmer **Sep 2021 - Jan 2023**

Java, Python, Angular, Spring Boot, FAISS, Docker, Kubernetes, AWS *Bangalore, India*

- Built a **document-intelligence platform** for **100K+ contracts daily** on **Docker/Kubernetes** (AWS EKS/ECS) and Jenkins CI/CD.
- Built a **semantic-search API** over **10M+ embeddings** (FAISS IndexIVFPQ), achieving **92% precision@5** at sub-200ms latency.
- Fine-tuned a **Legal-BERT model** for Named Entity Recognition, reaching **91% F1** and cutting manual contract review time **50%**.
- Built full-stack features for a **DevOps automation portal** (Angular, Spring Boot REST, Django reporting) and a Jenkins failure reporter.

VIGIL Lab, IIT Hyderabad | Research Intern, Computer Vision **Aug 2021 - Feb 2022**

Python, PyTorch, TensorFlow, Git, W&B *Hyderabad, India*

- Cut **object-detection** inference time **30%** (EfficientNet, YOLOv4; PyTorch, TensorFlow) via pruning and quantization.
- Built an experiment framework tracking **100+ runs** (Git, W&B) with automated hyperparameter tuning, improving mAP **12%**.

EDUCATION

Northeastern University | MS in Artificial Intelligence | Boston, MA **Jan 2023 - May 2025**

Head Teaching Assistant for NLP | Previously admitted to MS in ECE at CMU **GPA: 3.96/4**

Coursework: NLP, Deep Learning, LLMs, Generative AI, Reinforcement Learning, Software Design Patterns

NIT Arunachal Pradesh | B.Tech in Electronics & Communication | India **Aug 2016 - Jun 2020**

Coursework: Data Structures, Algorithms, Software Development, Computer Networks **Gold Medalist**